RABVPrime: An Automated Workflow for RABV Primer Design and Evaluation

Radasa Sathienkit[1], Sandeep Kasaragod[2,3], Kirstyn Brunker[2] and Joseph Hughes[3]

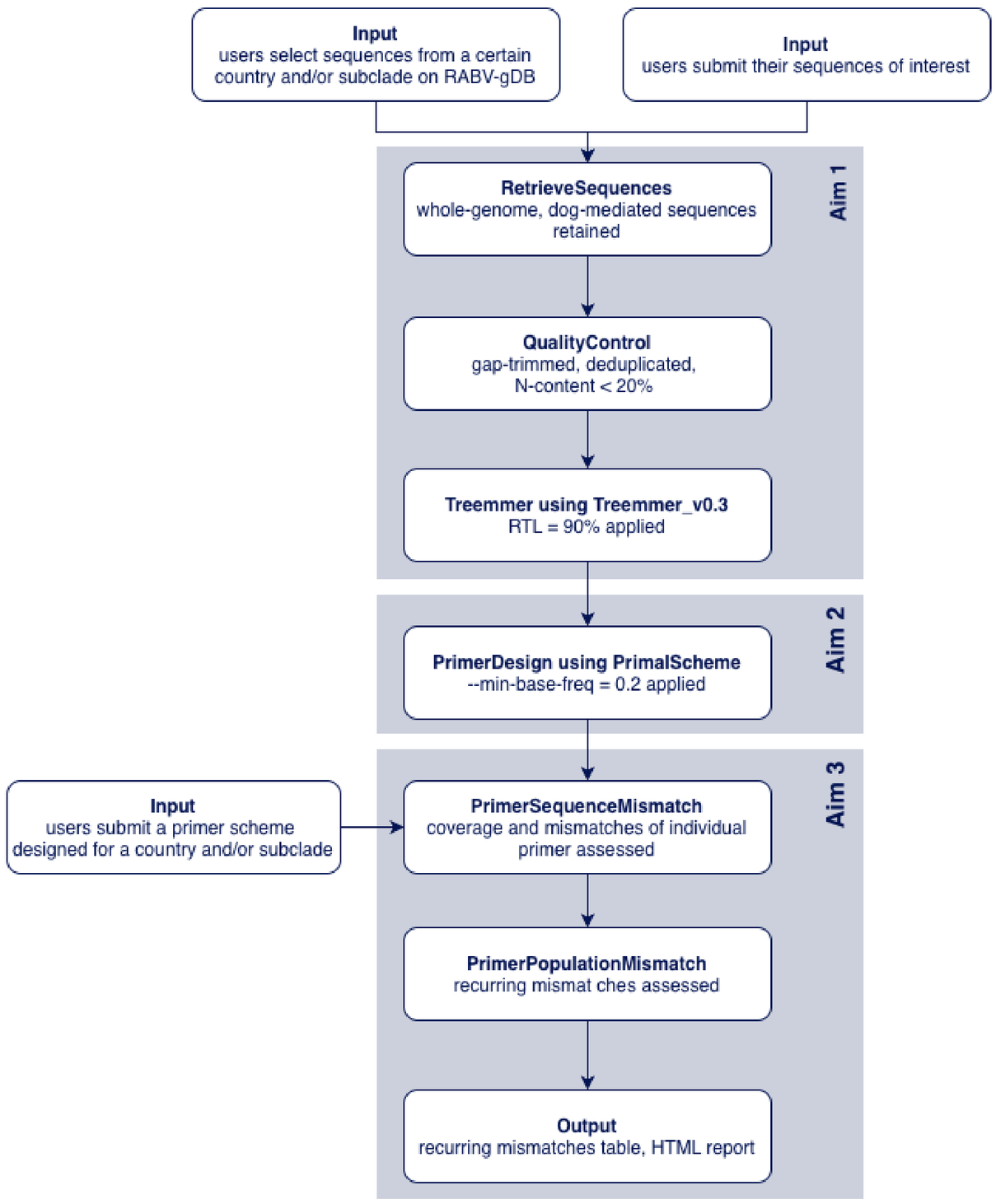
Introduction

- Genomic surveillance has been widely used to monitor circulating variants of pathogens such as SARS-CoV-2.
- Genomic surveillance is now also being used for rabies virus (RABV), the causative agent of rabies.
- Effective genomic surveillance relies on sequencing data, with primer-based approaches widely used for viral genome sequencing.
- Although primer design tools such as PrimalScheme are available, their application to RABV has several limitations. These limitations include difficulties in selecting representative sequences for primer design and interpreting the performance of designed primers without bioinformatics expertise.

Aims

- RABVPrime aims to address the limitations PrimalScheme has on RABV primer design and evaluation by:
    - Selecting high-quality, representative RABV sequences for primer design by PrimalScheme
    - Testing and validating different PrimalScheme parameters to identify suitable thresholds for the RABV genome
    - Evaluating primer–sequence mismatches and generating a user-friendly report that can be interpreted by users with and without bioinformatics expertise

Product



Evaluation

Figure 1. Heatmap showing potentially problematic primers for each sequence based on the default thresholds (≥1 3′ mismatches or ≥4 SNP/indel mismatches). The total number of mismatches for each primer is indicated.

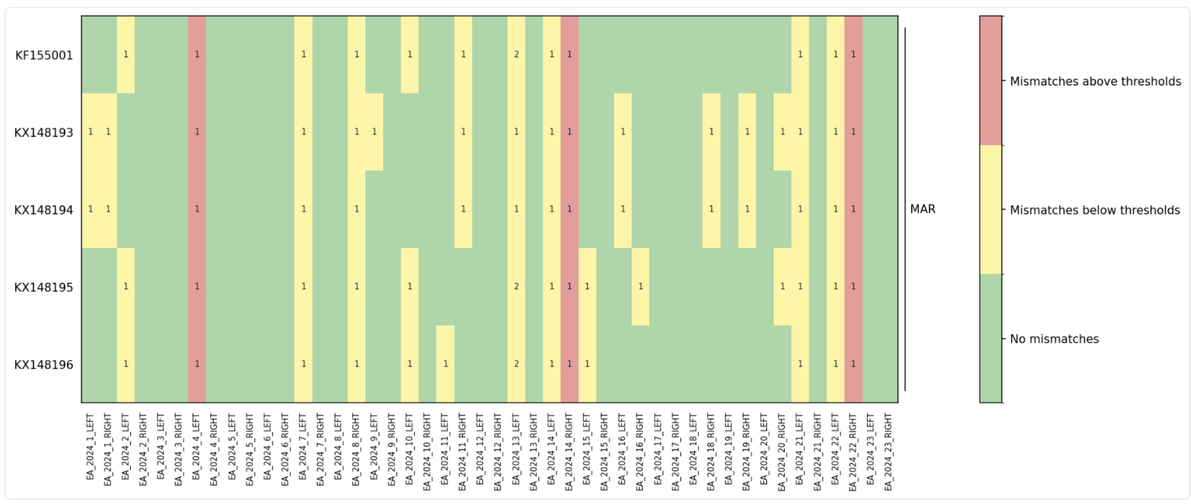


Table 1. Recurring mismatches observed across all sequences (mismatches at the same position with the same alternate base occurring in ≥20% of sequences).

Amplicon	Primer Name	Position in Primer	PrimerRef	PrimerAlt	Type	Count	Mismatch Proportion (%)
4	EA_2024_4_LEFT	17	G	A	3'	5/5	100
7	EA_2024_7_LEFT	18	G	A	snp	5/5	100
8	EA_2024_8_RIGHT	5	C	A	snp	5/5	100
13	EA_2024_13_LEFT	9	C	T	snp	5/5	100
14	EA_2024_14_LEFT	14	T	C	snp	5/5	100
14	EA_2024_14_RIGHT	24	T	G	3'	5/5	100
21	EA_2024_21_LEFT	5	C	T	snp	5/5	100
22	EA_2024_22_LEFT	20	A	C	snp	5/5	100

Primer Name	Issue	Recommendation
EA_2024_14_RIGHT	1 3' mismatch found in 100% of the sequences	Consider redesigning new primers
EA_2024_22_RIGHT	1 3' mismatch found in 100% of the sequences	
EA_2024_4_LEFT	1 3' mismatch found in 100% of the sequences	
EA_2024_13_LEFT	2 SNP mismatches found in at least 60% of the sequences	Consider alternate or degenerate primers
EA_2024_14_LEFT	1 SNP mismatch found in 100% of the sequences	
EA_2024_21_LEFT	1 SNP mismatch found in 100% of the sequences	
EA_2024_22_LEFT	1 SNP mismatch found in 100% of the sequences	

Conclusion and Further Work

- RABVPrime successfully addressed all aims.
- Sequencing data from *in vitro* experiments should be used to validate the accuracy of primer–sequence mismatch warnings.

1. College of Medical, Veterinary & Life Sciences, University of Glasgow, Glasgow, UK
2. School of Biodiversity, One Health & Veterinary Medicine, University of Glasgow, Glasgow, UK
3. MRC-University of Glasgow Centre for Virus Research, Glasgow, UK